CMF conservation as an arithmetic tool

A proof of $\mu(\zeta(2)) < 4.62675$ for the direction
$(10, 9, 11, -11; 2, 1, 3)$

Mathematical derivation with reproducible exact certificates

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Abstract

Conservation allows a matrix product to be decomposed into paths and blocks whose arithmetic is easier to see. Different initial rows and different primes may favor different certificates for the same endpoint. We explain this with an explicit 2×2 field, then use four long blocks of the native ${}_4F_3(1)$ field to derive an exact finite readout for the specified trajectory. A finite-field polynomial factorization proves the additional divisibility in all six required prime intervals. A Barnes integral identifies the error exactly and removes the dominant recurrence mode; an elementary cone argument then gives the sharp exponential upper rate. A nonzero observability determinant supplies independence at three consecutive depths. These ingredients give $\mu(\zeta(2)) \leq 62636575/13537936 < 4.62675$. The finite chamber and numerical constant certificates use exact rational arithmetic. No formal proof assistant is used.

1 The idea: one endpoint, several arithmetic descriptions

An arithmetic estimate need not use the same path for every initial row or every prime. Conservation makes the endpoint product independent of the admissible path, so each valid estimate constrains the same rational vector. A favorable first block may preserve an eigenline or a congruence lattice for one row, while another row benefits from another block order.

For the present trajectory, the effective implementation is a fixed four-block route followed by prime-dependent evaluation of its exact readout. The last $11N$ moves are treated as one finite-difference operator. Their cancellation survives reduction modulo a prime and produces the extra common divisor. Estimating these moves one at a time would discard the polynomial degree reduction which proves that divisor.

1.1 Conservation changes the available arithmetic certificates

Use row vectors and multiply on the right. A conservative matrix field obeys

$$M_x(w)M_y(w + e_x) = M_y(w)M_x(w + e_y).$$

Thus two admissible paths related by elementary square exchanges have the same endpoint product. The evaluation points move when the order changes: this is not permission to commute two matrices evaluated at the same point. All exchanges must stay within the domain where the required matrices and identities are defined.

For a prime ℓ, let $c_\ell(V)$ denote the minimum ℓ-adic valuation of the entries of a rational matrix or row V, with $v_\ell(0) = +\infty$. The elementary valuation inequality gives

$$c_\ell(UV) \geq c_\ell(U) + c_\ell(V).$$

Indeed, every summand in every entry of UV has valuation at least the right-hand side, and addition cannot decrease that lower bound.

Suppose a path γ is decomposed into full blocks $B_{\gamma,1}, \dots, B_{\gamma,k}$, and let its endpoint product be W. For an initial row r, a useful certificate is

$$L_{\ell,r}(\gamma) = c_\ell(rB_{\gamma,1}) + \sum_{j=2}^{k} c_\ell(B_{\gamma,j}), \qquad c_\ell(rW) \geq L_{\ell,r}(\gamma).$$

More accurate certificates retain the transported row through later blocks, or retain a lattice of possible rows. The blocks can contain any number of elementary steps. Replacing a block by separate entrywise estimates can discard cancellations, since the displayed product inequality can be strict.

If $\mathcal{P}$ is any family of admissible paths to the same endpoint, conservation therefore gives

$$c_\ell(rW) \geq \max_{\gamma \in \mathcal{P}} L_{\ell,r}(\gamma).$$

The maximizing path can depend on r, on ℓ, and on the endpoint parameter N. These choices are compatible because every certificate concerns the same rational endpoint vector. They do not change the approximant or its actual reduced denominator.

For several initial rows, the useful common-content bound is

$$\min_r \max_{\gamma \in \mathcal{P}} L_{\ell,r}(\gamma) \;\geq\; \max_{\gamma \in \mathcal{P}} \min_r L_{\ell,r}(\gamma).$$

The left side allows a different path for each initial row. The inequality can be strict even for a 2×2 field and one elementary square.

1.2 An exact toy example with both kinds of dependence

Consider the rational potential

$$F(x,y) = \begin{pmatrix} 1 + x + 2y + 2xy & 3y(1-x) \\ 3x(1-y) & 1 + 2x + y + 2xy \end{pmatrix}, \qquad M_i(w) = F(w)^{-1} F(w + e_i).$$

This defines a rational conservative field on its nonsingular domain: both products around a square telescope to $F(w)^{-1}F(w + e_x + e_y)$. The four corners used below are nonsingular. Writing X for an x-step and Y for a y-step, their matrices are

$$A = M_x(0,0) = \begin{pmatrix} 2 & 0 \\ 3 & 3 \end{pmatrix}, \quad B = M_y(0,0) = \begin{pmatrix} 3 & 3 \\ 0 & 2 \end{pmatrix},$$

$$C = M_y(1,0) = \begin{pmatrix} 3 & 0 \\ -3 & 2 \end{pmatrix}, \quad D = M_x(0,1) = \begin{pmatrix} 2 & -3 \\ 0 & 3 \end{pmatrix}.$$

Direct multiplication gives

$$W = AC = BD = 6I_2, \qquad AB \neq BA.$$

The first coordinate row $e_1 = (1,0)$ is a left eigenvector of A with eigenvalue 2; the second coordinate row $e_2 = (0,1)$ is a left eigenvector of B with the same eigenvalue. In contrast,

$$e_1 B = (3,3), \qquad e_2 A = (3,3).$$

All four whole matrices have content zero at both primes 2 and 3. Consequently a bound using only their individual matrix contents certifies no positive content. Retaining the first transported row gives instead:

initial row	ℓ	$L_{\ell,r}(XY)$	$L_{\ell,r}(YX)$	best order
e_1	2	1	0	XY
e_2	2	0	1	YX
e_1	3	0	1	YX
e_2	3	1	0	XY

For $\ell = 2$, send e_1 along XY and e_2 along YX. This certifies that both endpoint rows are divisible by 2. For $\ell = 3$, reverse those choices; both are divisible by 3. Since the endpoint vectors agree exactly along the two paths, both endpoint rows are divisible by 6. Either single order used for both rows gives only the common-content lower bound zero at each prime under this particular certificate. The endpoint calculation $W = 6I_2$ independently confirms that the optimized bounds are sharp.

This example is deliberately small enough that the endpoint product can be inspected directly. Its purpose is to exhibit exactly the information lost by an arithmetic estimate, and how conservation recovers that information. For large hypergeometric products one seeks symbolic block identities that supply the same certificates without expanding the full endpoint matrix.

The same effect as denominator clearing. There is also a rational version of the square with edge matrices $A/6, B/6, C, D$, so both endpoint products are I_2. It is realized by bilinearly interpolating potential values $I_2, A/6, B/6, I_2$ at the four corners and again using $M_i(w) = F(w)^{-1}F(w + e_i)$. For e_1, the XY calculation clears denominators by 3, whereas the YX calculation clears them by 2. For e_2 the roles reverse. Since either calculation concerns the same endpoint row, localizing at each prime certifies integrality: use the path with no 2-denominator at 2 and the path with no 3-denominator at 3. Neither whole-matrix denominator estimate alone displays this fact.

Equivalently, if the positive integers D_γ all clear denominators of the same rational row, then $\gcd_\gamma D_\gamma$ clears them as well, by Bézout's identity. For two rows, independently optimizing their certificates gives the common denominator

$$\operatorname{lcm}\left(\gcd_\gamma D_{1,\gamma},\ \gcd_\gamma D_{2,\gamma}\right).$$

This divides the less flexible expression

$$\gcd_\gamma \operatorname{lcm}(D_{1,\gamma}, D_{2,\gamma}).$$

The rational toy square gives 1 for the first expression and 6 for the second. Prime-specific ordering is precisely the local form of taking these greatest common divisors; it does not splice different approximants together.

What an arithmetic proof must record. A usable certificate specifies the initial row, the prime range, the entire block order, every block's starting point, and the retained transported row or lattice. Numerical endpoint readouts suggest such certificates; a uniform identity or valuation argument must establish them for all admissible parameters. A finite set of successful readouts alone is not that uniform argument.

2 Comparison with the literature

Rhin and Viola's permutation-group method compares representations of the same linear form obtained from transformations of a multiple integral. Factorial quotients in the transformation formulas supply additional p-adic information; their 1996 paper obtains $\mu(\zeta(2)) < 5.441243$ [1]. Its arithmetic analysis is particularly explicit in Sections 3–4. This provides the relevant analogy: several equivalent descriptions can certify more divisibility than one description considered in isolation.

The choice of description separately at each prime has a direct precedent. In Zudilin's hypergeometric treatment, Lemma 7 maximizes valuations of factorial quotients over parameter permutations. Section 6 then compares two different hypergeometric descriptions of the same pair of coefficients and takes the pointwise maximum of their arithmetic profiles, $\widetilde{\varphi}(x) = \max\{\varphi(x), \widehat{\varphi}(x)\}$. Equations (28)–(29), followed by the asymptotic calculation in Section 7, yield $\mu(\zeta(2)) \le 5.09541178\ldots$ [2]. Thus prime-dependent selection should be presented as a familiar arithmetic principle, here organized through CMF path identities.

CMFs also already have an arithmetic role in the literature. David's potential matrix is independent of the lattice path (Definition 11). Lemma 36 and Corollary 37 use identities relating horizontal and vertical directions to establish factorial divisibility for a $\zeta(3)$ field [3]. The July 2026 revision of Weinbaum et al. studies CMF asymptotics through a discrete Levinson-type framework and explicitly separates structural results from experimental arithmetic conjectures [4].

In the present proposal, the representations are shifted matrix products along admissible paths in a known conserving family. A selected row is transported exactly into each block, so a certificate can retain congruence information lost by an entrywise matrix estimate. One may then optimize the certificate for each row and prime, while conservation keeps the endpoint readout fixed. This is the concrete CMF mechanism to investigate; the extra prime intervals for the present $\zeta(2)$ trajectory are proved by the uniform polynomial argument in Appendix D. Suitable descriptions of the classical work are "technically intricate", "case-intensive", and "dependent on carefully chosen transformations".

3 The specified $\zeta(2)$ trajectory

We use the native field $\mathrm{pFq}(4,3,1)$, with initial vertex and direction

$$w_0 = (1,1,1,0;1,1,1), \qquad v = (10,9,11,-11;2,1,3).$$

Let $W(N)$ be its unscaled endpoint product, acting on rows on the right, and set

$$R = \begin{pmatrix} 0 & 2 & -1 \\ 2 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} p_N^{\mathrm{raw}} \\ q_N^{\mathrm{raw}} \end{pmatrix} = RW(N)e_0.$$

Thus $p_N^{\mathrm{raw}}/q_N^{\mathrm{raw}}$ is exactly the expression in the experiment, $(2W_{10} - W_{20})/(2W_{00})$. The column labels below are $r = 0, 1, 2$. The two initial rows of R are transported separately when deriving arithmetic bounds; they are not reset at the boundaries of blocks.

At a vertex $w = (x_0, x_1, x_2, -m; y_0, y_1, y_2)$, write

$$Q_w(s) = s\prod_{i=0}^{2}(s + y_i - 1) - (s - m)\prod_{i=0}^{2}(s + x_i) = \sum_{j=0}^{3} a_j s^j, \quad T_w = \begin{pmatrix} 0 & 0 & -a_0/a_3 \\ 1 & 0 & -a_1/a_3 \\ 0 & 1 & -a_2/a_3 \end{pmatrix}.$$

An upper increase has matrix $I + T_w/x_i$. A lower increase is the inverse of $I + T_{w+e_{y_i}}/y_i$, evaluated at the target in T. A decrease of x_3 is likewise the inverse upper move evaluated at its target. These evaluation points are part of the multiplication order.

3.1 The four chosen blocks

For $t = 3$ or 4, the following blocks reach the same endpoint:

block	direction per block step	number of steps
A	$(1,1,1,0;0,0,0)$	tN
B	$(0,0,0,0;2,1,3)$	N
C	$(10-t, 9-t, 11-t, 0; 0, 0, 0)$	N
D	$(0,0,0,-1;0,0,0)$	$11N$

Within each step, make its coordinate moves from left to right, always evaluating at the current vertex. The first block raises all positive uppers enough that the lower moves are regular. The lower block then establishes the harmonic shifts; the remaining upper block establishes the binomial widths; the final negative-upper block performs a long finite difference. In particular this decomposition treats the last $11N$ steps together, rather than estimating them independently.

At the endpoint all three paired widths $x_i - y_i$ equal $8N$. This is the reason that the readout below contains three binomial polynomials of a common width. Appendix C records the complete $t = 3$ vertices and compatible integral normalizations, including every block boundary. Appendix A verifies all generators and the regularity of both routes uniformly in N.

3.2 Reading the arithmetic from the rows

Put $d_n = \operatorname{lcm}(1, \dots, n)$, with $d_0 = 1$, and let $H_n^{(2)} = \sum_{k=1}^{n} k^{-2}$, with $H_0^{(2)} = 0$. Define

$$
\begin{aligned}
C_N &= \binom{10N}{2N}\binom{9N}{N}\binom{11N}{3N}, \qquad m = 11N, \\
F(s) &= \binom{s+10N}{8N}\binom{s+9N}{8N}\binom{s+11N}{8N}, \qquad c_j = (-1)^j \binom{m}{j} F(j), \\
S_r(s) &= \sum_{j=0}^{m} (-1)^j \binom{m}{j} \frac{s^r F(s) - j^r F(j)}{s - j}, \\
J_\eta(P) &= \sum_{h=0}^{\deg P} \frac{(-1)^h}{h+1} \Delta^h P(-\eta), \qquad \Delta P(s) = P(s+1) - P(s).
\end{aligned}
$$

We use $0^0 = 1$ and the polynomial value of each divided difference at a removable singularity. The exact native readout is

$$
C_N RW(N) e_r = 2 \begin{pmatrix} \displaystyle\sum_{j=0}^{m} j^r c_j H_{j+\eta}^{(2)} + J_\eta(S_r) \\ \displaystyle\sum_{j=0}^{m} j^r c_j \end{pmatrix}, \qquad 2N \le \eta \le 9N. \tag{1}
$$

This is a uniform identity, proved in Appendix A, and its value is independent of the admissible integer η. The first row contains the harmonic and polynomial correction terms; the second is an integer binomial sum after multiplication by C_N. The polynomial degree drops from $24N + r - 1$ to at most $13N + r - 1$ because a finite difference of order $11N$ kills all lower powers. That is a whole-block cancellation.

The matrix readout therefore does more than suggest a growth rate. Its exact expression supplies the integer-valued polynomial basis, the possible denominator indices, and the binomial carries that can force common divisors. A raw list of rational approximants would hide this structure. Discovering a conserving family provides a systematic route to the identities that expose it.

4 The arithmetic normalization

For real x, y, set

$$\begin{aligned}\rho(x,y) &= \lfloor 11x\rfloor - \lfloor y\rfloor - \lfloor 11x-y\rfloor \\ &\quad + \lfloor 10x+y\rfloor + \lfloor 9x+y\rfloor + \lfloor 11x+y\rfloor - 3\lfloor 8x\rfloor \\ &\quad - \lfloor 2x+y\rfloor - \lfloor x+y\rfloor - \lfloor 3x+y\rfloor,\\ \phi(x) &= \min_{0\le y<1}\rho(x,y), \qquad G_N = \prod_{\substack{\sqrt{50N}<p\le 13N\\ p \text{ prime}}} p^{\phi(N/p)}, \qquad D_{N,r} = d_{12N}d_{13N+r}.\end{aligned}$$

Each summand in the carry interpretation of ρ is nonnegative. The large-prime cutoff ensures that only the first level of Legendre's factorial valuation formula is involved.

Appendix B proves the baseline statement

$$\frac{C_N D_{N,r}}{G_N} RW(N)e_r \in \mathbb{Z}^2 \quad (N\ge 1,\ r=0,1,2). \tag{2}$$

It does so by bounding the divided differences in an integer-valued binomial basis, accounting for a missing harmonic prime by a binomial carry, and transferring common divisibility through Newton coefficients. Thus this is a proof of the arithmetic of the actual native rows, not an extrapolation from the checked endpoints.

The additional common divisor comes from the following six prime intervals. Let $H_{N,r}$ be the product of primes $p > \sqrt{50N}$ belonging to one of the following *open* intervals:

$$\begin{aligned}&((13N+r)/2, 8N), \quad ((13N+r)/3, 9N/2), \quad ((13N+r)/4, 10N/3),\\ &((13N+r)/5, 8N/3), \quad ((13N+r)/6, 9N/4), \quad ((13N+r)/12, 9N/8).\end{aligned} \tag{3}$$

Theorem 1 (Arithmetic of the native readouts)**.** *For every $N \ge 1$ and $r = 0, 1, 2$,*

$$\boxed{\frac{C_N d_{12N} d_{13N+r}}{G_N H_{N,r}} RW(N)e_r \in \mathbb{Z}^2.}$$

Proof. The baseline argument in Appendix B clears all denominators and divides out G_N. It remains to establish one further factor of each prime in the six open intervals. Appendix D proves this simultaneously for all six intervals and all three columns. Here is the mechanism, before its detailed calculation.

Write $8N = ap + b$, $11N = Mp + d$, with $0 \le b, d < p$, and let $h = \phi(N/p)$. In $\mathbb{F}_p[X]$ form

$$\mathcal{Q}(X) = \frac{\displaystyle\prod_{u=9}^{11} \binom{X+uN}{b}}{(X^p - X)^{h-1} \displaystyle\prod_{j=0}^{d}(X-j)}.$$

For $h = 0$, the expression means multiplication by $X^p - X$ in the numerator. The carry lower bound proves that this is a polynomial: every denominator root occurs in the numerator with at least the required multiplicity. In a shell labeled by $K = 2, 3, 4, 5, 6, 12$,

$$\deg(X^r \mathcal{Q}) = 13N + r - (K-1)p - 1 < p - 1.$$

Therefore $\sum_{k\in\mathbb{F}_p} k^r \mathcal{Q}(k) = 0$.

The whole-block rational kernel, reduced at $s = k + pV$, separates into $\mathcal{Q}(k)$ times a rational function of the high digit V. The floor-cell table shows that the latter function is independent of k on the support of $\mathcal{Q}$. Choose the admissible shift $\eta = p$ or $2p$. Both the leading harmonic residue and the leading Newton-coefficient residue then contain the same vanishing sum. Both parts of the numerator gain a factor of p, and the denominator already has the required factor. Combining the distinct primes proves the assertion. The appendix accounts for derivative nodes and for the carry chambers where a termwise argument would fail. □

This identifies precisely how the conserving family helps the proof. It transports the original rows to a useful four-block description; the exact readout records a long finite difference; and its polynomial structure supplies the uniform arithmetic cancellation. The choice of η here is a prime-dependent representation of that same readout. The toy example separately illustrates literal prime-dependent path orders. We do not identify these two freedoms without an additional path identity.

5 The exponential cost of the arithmetic normalization

Define the arithmetic rate

$$\gamma = \int_{1/13}^{\infty} \frac{\phi(x)}{x^2}\, dx.$$

The function ϕ is bounded and one-periodic, and vanishes on $(0, 1/13)$. The prime number theorem, applied on its continuity intervals, gives $N^{-1} \log G_N \longrightarrow \gamma$. For completeness, first restrict to $p \geq N/K$, where there are only finitely many intervals, then bound the omitted part by $\|\phi\|_\infty \vartheta(N/K)/N = O(1/K)$. The excluded primes $p \leq \sqrt{50N}$ contribute $o(N)$ and do not affect the limit. Here $\vartheta(x) = \sum_{p \leq x} \log p$.

The six interval lengths in (3), divided by N, sum to

$$\left(8 - \frac{13}{2}\right) + \left(\frac{9}{2} - \frac{13}{3}\right) + \left(\frac{10}{3} - \frac{13}{4}\right) + \left(\frac{8}{3} - \frac{13}{5}\right) + \left(\frac{9}{4} - \frac{13}{6}\right) + \left(\frac{9}{8} - \frac{13}{12}\right) = \frac{233}{120}.$$

Consequently Theorem 1 gives the arithmetic exponent

$$c = 25 - \gamma - \frac{233}{120} < 10.929906.$$

The strict numerical comparison uses a directed rational enclosure, not a floating-point integral; Appendix E explains it.

In particular, with $k_N = d_{12N} d_{13N}/(G_N H_{N,0})$, we have

$$\lim_{N \to \infty} \frac{\log k_N}{N} = 25 - \gamma - \frac{233}{120} < 10.929906.$$

Only a subexponential factor is lost by omitting primes below $\sqrt{50N}$ or by using open interval endpoints.

6 Analytic estimates and nonvanishing from the native CMF

Let W_N be the native endpoint product, and write

$$\mathsf{P}_{N,r} = (RW_N)_{0r}, \qquad \mathsf{Q}_{N,r} = (RW_N)_{1r}, \qquad \mathsf{E}_{N,r} = \zeta(2)\mathsf{Q}_{N,r} - \mathsf{P}_{N,r} \quad (0 \leq r \leq 2).$$

These symbols denote the rational matrix readouts, before integer normalization. Put

$$F_N(s) = \prod_{u=9}^{11} \binom{s+uN}{8N}, \quad C_N = F_N(0), \quad m = 11N, \quad L_m(s) = \frac{(-1)^m m!}{s(s-1)\cdots(s-m)}.$$

The finite formula provides the following exact integral identification, not merely a numerically identified limit.

6.1 The Barnes integral identifies the error exactly

For $\eta \geq 0$ and a function for which the integral converges, set

$$\mathcal{I}_\eta(g) = \frac{1}{2\pi i} \int_{-\eta-1/2-i\infty}^{-\eta-1/2+i\infty} \left(\frac{\pi}{\sin \pi s}\right)^2 g(s)\, ds.$$

For every polynomial P,

$$\mathcal{I}_\eta(P) = J_\eta(P).$$

Here is a direct verification. Moving the line one unit to the right crosses one double pole, whose residue is $P'(-\eta)$; the horizontal segments tend to zero because of the exponential decay of the kernel. Thus the integral functional obeys $\mathcal{I}_\eta(P(s+1) - P(s)) = P'(-\eta)$. This determines a linear functional on all polynomials, since the forward-difference map on polynomials is onto. It agrees with the binomial-basis identity defining J_η.

For $j \geq 0$, closing a large half-integer rectangle to the right gives

$$\mathcal{I}_\eta\left(\frac{1}{s-j}\right) = H^{(2)}_{j+\eta} - \zeta(2).$$

Indeed, the residue at an integer $k \neq j$ is $-(k-j)^{-2}$, and the residue at $k = j$ is $\pi^2/3 = 2\zeta(2)$. The integral on the distant right edge tends to zero. The left edge is oriented upward, which accounts for the minus sign in the residue sum.

The exact partial-fraction identity

$$s^r F_N(s) L_m(s) = S_r(s) + \sum_{j=0}^{m} \frac{j^r (-1)^j \binom{m}{j} F_N(j)}{s-j}$$

therefore yields

$$\boxed{C_N \mathsf{E}_{N,r} = -2\mathcal{I}_\eta(s^r F_N L_m).} \tag{A1}$$

Thus the constant in the error is exactly $\zeta(2)$. Take $\eta = 6N$. The line may be moved from $\Re s = -6N - 1/2$ to $\Re s = -6N$: the only integer crossed is a zero of F_N of multiplicity three, so it cancels the double pole of the sine kernel.

6.2 A coarse exponential bound removes the dominant mode

Use $t = -s$ and the reflection formula for the gamma function. Since

$$L_m(-t) = -B(t, m+1),$$

and the three complementary shifts are $N, 2N, 3N$, whose sum is even, (A1) becomes

$$\frac{C_N \mathsf{E}_{N,r}}{2} = \frac{1}{2\pi i} \int_{6N-i\infty}^{6N+i\infty} (-t)^r B(t, 11N+1) \frac{\prod_{v=1}^{3} \Gamma(t - vN)\Gamma((v+8)N + 1 - t)}{\Gamma(8N+1)^3} \frac{\sin \pi t}{\pi}\, dt. \tag{A2}$$

All gamma arguments in this formula have positive real part along the line. For $a > 0$, real y, and $0 < \theta < \pi/2$, rotation of the Euler integral by $\operatorname{sgn}(y)\theta$ proves

$$|\Gamma(a + iy)| \le \Gamma(a)(\cos\theta)^{-a} e^{-\theta|y|}.$$

The rotation is justified by exponential decay on the outer arc and $a > 0$ on the inner arc. Also, directly from the beta integral, $|B(6N + iy, 11N + 1)| \le B(6N, 11N + 1)$.

The six gamma arguments in (A2), at $t = 6N+iy$, have real parts $3N, 3N+1, 4N, 4N+1, 5N, 5N+1$. Choose $\theta = \pi/4$. Their combined bound supplies $2^{12N+3/2} e^{-3\pi|y|/2}$; the sine contributes at most $e^{\pi|y|}/2$. Consequently the remaining integral is bounded by a constant times N^r, uniformly for $N \ge 1$. More explicitly,

$$|C_N \mathsf{E}_{N,r}| \le K_r N^r 2^{12N} B(6N, 11N + 1) \frac{\prod_{k=3}^{5} \Gamma(kN)\Gamma(kN + 1)}{\Gamma(8N + 1)^3}, \tag{A3}$$

where K_r is independent of N. Stirling's formula gives

$$\limsup_{N\to\infty} \frac{1}{N} \log |C_N \mathsf{E}_{N,r}| \le h,$$

$$h = 12\log 2 + 6\log 6 + 11\log 11 - 17\log 17 + 2\sum_{k=3}^{5} k\log k - 24\log 8 < -18. \tag{A4}$$

Since $C_N \ge 1$, this already proves exponential decay of the entire unscaled error row. It is deliberately a coarse bound; it will identify which invariant modes are present, after which the matrix recurrence supplies the sharp upper rate.

6.3 The limiting matrix and an elementary cone argument

Let M_N be the native product of the 47 unit moves from the endpoint at N to that at $N + 1$. Conservation gives $W_{N+1} = W_N M_N$. For $N \ge 1$ put

$$S_N = \operatorname{diag}(1, N^{-1}, N^{-2}), \qquad A_N = S_N^{-1} M_N S_{N+1}.$$

Every entry is a rational function of N, and direct degree comparison in the native companion matrix gives $A_N = A_\infty + O(N^{-1})$, where

$$T = \begin{pmatrix} 0 & 0 & 10890/13 \\ 1 & 0 & 2305/13 \\ 0 & 1 & 42/13 \end{pmatrix},$$

$$A_\infty = \prod_{u\in\{10,9,11\}} (I + T/u)^u (I - T/11)^{-11} \prod_{v\in\{2,1,3\}} (I + T/v)^{-v}. \tag{A5}$$

To see the limit directly, each unit move changes the vertex by only a bounded amount, while $S_N^{-1} T_w S_N / N \to T$ and its parameter denominator divided by N tends to its displayed slope. All the limiting factors are invertible. Since they are rational functions of the same matrix T, their order in (A5) is immaterial.

Write the characteristic polynomial as $z^3 + c_2 z^2 + c_1 z + c_0$. Exact rational arithmetic gives

$$c_0 = \frac{1855425871872}{895833302668638834554683750087890625},$$

$$c_1 = \frac{611991758100915534539656279385190 4}{32000061404626447809127858236889541015625},$$
$$c_2 = \frac{266957778538934989798238376160038092809}{153423248579285748095495531 25}.$$

Its discriminant is strictly negative. Thus there is one real root $-\Lambda$ and a nonreal conjugate pair with common modulus ρ, where

$$17400086428 < \Lambda < 17400086429, \qquad \rho^2 = \frac{c_0}{\Lambda} < 1. \tag{A6}$$

In particular, all three roots are distinct, and A_∞ is complex-diagonalizable.

For completeness, the needed recurrence conclusion has an elementary proof. In a fixed eigenbasis split a row solution of $x_{N+1} = x_N A_N$ into its dominant coordinate u_N and remaining pair v_N, using the maximum norm on the pair. For any sufficiently small $\varepsilon > 0$, the perturbation from the diagonal limiting matrix has norm at most ε for all sufficiently large N. The cone $|u_N| \geq \|v_N\|$ is then forward invariant, and inside it

$$|u_{N+1}| \geq (\Lambda - 2\varepsilon)|u_N|, \qquad \|v_{N+1}\| \leq (\rho + 2\varepsilon)|u_N|.$$

Choose $\Lambda - 2\varepsilon > \max(1, \rho + 2\varepsilon)$. A nonzero solution tending to zero cannot enter this cone. Outside it, $|u_N| < \|v_N\|$ at every sufficiently large index, so

$$\|v_{N+1}\| \leq (\rho + 2\varepsilon)\|v_N\|.$$

Letting ε decrease to zero proves a growth upper rate $\log \rho$ for every decaying solution. Without the decay condition, the ordinary operator-norm bound gives upper rate $\log \Lambda$. The zero solution satisfies both upper bounds automatically.

Apply this first to the fixed error row $(\zeta(2)R_{1,*} - R_{0,*})W_N S_N$, which tends to zero by (A3), and then to either fixed row of $RW_N S_N$. Removing S_N changes only polynomial factors. We obtain

$$\limsup_N \frac{1}{N} \log |\mathsf{E}_{N,r}| \leq \log \rho, \qquad \limsup_N \frac{1}{N} \log |\mathsf{Q}_{N,r}| \leq \log \Lambda. \tag{A7}$$

These are uniform exponential upper estimates, and do not require a lower estimate for an oscillatory error.

6.4 Certified numerical rates

Stirling's formula gives

$$\lim_N \frac{\log C_N}{N} = c_C = 10 \log 10 + 9 \log 9 + 11 \log 11 - 2 \log 2 - 3 \log 3 - 24 \log 8.$$

The accompanying standard-library program verifies by exact rational arithmetic that

$$c_C + \log \Lambda < 38.168733, \qquad c_C + \log \rho < -24.467842. \tag{A8}$$

For reproducibility, logarithms are enclosed by scaling a positive rational argument into $[1, 2]$ and using

$$\log x = 2 \sum_{k=0}^{K-1} \frac{z^{2k+1}}{2k+1} + R_K, \quad z = \frac{x-1}{x+1}, \quad 0 \leq R_K \leq \frac{2z^{2K+1}}{(2K+1)(1-z^2)}.$$

The implementation uses $K = 32$, handles signs of all coefficients when combining intervals, checks the cubic discriminant and the root bracket (A6), and uses $\log \rho = (\log c_0 - \log \Lambda)/2$. No floating-point value is used in its assertions.

6.5 A bounded gap of at most two for column zero

The product W_N is invertible on the chosen regular paths. The exact observability determinant is

$$\det(e_0, A_\infty e_0, A_\infty^2 e_0) = \frac{7214325933922030302580492752459309064550608447544202530775113660364 8}{2846959945518179368207403132258698202971898733418386758442993164062 5} \neq 0. \tag{A9}$$

Thus $(e_0, A_N e_0, A_N A_{N+1} e_0)$ is invertible for all sufficiently large N. Given integers a, b with $b > 0$, the row

$$(bR_{0,*} - aR_{1,*})W_N S_N = (-2a, 2b, -b)W_N S_N$$

is nonzero. It cannot annihilate all three columns in that matrix. Therefore at least one of

$$b\mathsf{P}_{N,0} - a\mathsf{Q}_{N,0}, \quad b\mathsf{P}_{N+1,0} - a\mathsf{Q}_{N+1,0}, \quad b\mathsf{P}_{N+2,0} - a\mathsf{Q}_{N+2,0}$$

is nonzero. This supplies the needed nonvanishing with a bounded gap, directly from the native three-dimensional matrices. Multiplying each pair by any nonzero normalization preserves this property.

6.6 The irrationality-measure conclusion

Theorem 1 and the preceding rate calculation supply nonzero scalars k_N with

$$P_N = C_N k_N \mathsf{P}_{N,0} \in \mathbb{Z}, \qquad Q_N = C_N k_N \mathsf{Q}_{N,0} \in \mathbb{Z}, \qquad \limsup_N \frac{\log |k_N|}{N} \leq 10.929906. \tag{A10}$$

Here $k_N = d_{12N} d_{13N}/(G_N H_{N,0})$. Equations (A7)–(A8) then give

$$\limsup_N \frac{\log |Q_N|}{N} \leq \sigma = 49.098639, \qquad \limsup_N \frac{\log |Q_N \zeta(2) - P_N|}{N} \leq -\tau = -13.537936.$$

Here is the irrationality-measure step explicitly. For any small $\varepsilon > 0$, take n just large enough that $be^{-(\tau-\varepsilon)n} < 1/2$. At one of $N = n, n+1, n+2$, (A9) ensures the integer $bP_N - aQ_N$ is nonzero. Hence

$$1 \leq |bP_N - aQ_N| \leq b|Q_N| \left|\zeta(2) - \frac{a}{b}\right| + b|Q_N \zeta(2) - P_N|,$$

which gives a positive constant times $b^{-1-(\sigma+\varepsilon)/(\tau-\varepsilon)}$ as a lower bound for the rational-approximation error. Letting $\varepsilon \downarrow 0$ proves

$$\boxed{\mu(\zeta(2)) \leq 1 + \frac{\sigma}{\tau} = \frac{62636575}{13537936} < 4.62675.}$$

All ingredients have now been supplied: the finite CMF identity, the uniform arithmetic normalization, the analytic upper estimates, and bounded-gap independence. Here $\mu(\alpha)$ is the usual irrationality measure: the infimum of μ such that for every $\varepsilon > 0$ only finitely many reduced a/b satisfy $|\alpha - a/b| < b^{-\mu-\varepsilon}$.

7 The role of the exact code

The proof reduces its finite checks to explicit rational identities and finite arrangements. The accompanying Python package verifies the 390 floor cells in the eight-row chamber table, computes the rational limiting matrix and its discriminant, brackets its real eigenvalue, checks the observability determinant, and encloses the logarithms with positive series and explicit tails. Its carry-integral computation also uses a finite rational partition and a proved tail bound. These finite certificates check the stated reductions for all parameters covered by each cell; they are not tests at a chosen finite list of depths.

Other scripts provide additional diagnostics: RamanujanTools checks the native matrices against the four-block products, the finite-functional checker checks general vertices, and independent integer arithmetic checks the local kernel formulas. These diagnostics help catch mistakes but are not the logical justification for universal statements.

The uniform arithmetic argument keeps the cancellation of the entire rational kernel. Each finite certificate is used only after its reduction to a finite set of rational identities or floor cells has been established in the text.

A Uniform derivation of the finite readout

We use row vectors and the unscaled native generators. Assume

$$x_i, y_i \in \mathbb{Z}, \quad x_i \ge y_i \ge 1 \quad (0 \le i \le 2), \qquad m \in \mathbb{Z}_{\ge 0},$$

and put $b_i = x_i - y_i$ and

$$I_w = \{\eta \in \mathbb{Z} : \operatorname{med}(y_0 - 1, y_1 - 1, y_2 - 1) \le \eta \le \min_i (x_i - 1)\}.$$

Assume $I_w \neq \varnothing$. Define

$$F(s) = \prod_{i=0}^{2} \binom{s + x_i - 1}{b_i}, \quad C = F(0) > 0, \quad \omega_j = (-1)^j \binom{m}{j}, \quad c_j = \omega_j F(j),$$

$$A(s) = s \prod_i (s + y_i - 1), \qquad D(s) = (s - m) \prod_i (s + x_i), \qquad Q(s) = A(s) - D(s).$$

The two monic quartic terms cancel. Write $Q(s) = a_0 + a_1 s + a_2 s^2 + a_3 s^3$ and assume $a_3 \neq 0$. For a polynomial f, set

$$S_f(s) = \sum_{j=0}^{m} \omega_j \frac{f(s)F(s) - f(j)F(j)}{s - j}, \qquad \ell_\eta(f) = \begin{pmatrix} \sum_{j=0}^m f(j) c_j H^{(2)}_{j+\eta} + J_\eta(S_f) \\ \sum_{j=0}^m f(j) c_j \end{pmatrix},$$

where $J_\eta(\binom{s+\eta}{h}) = (-1)^h/(h+1)$. All the divided differences defining S_f are polynomials.

1. The two identities for the finite functional. Let P be any polynomial. Pascal's identity and

$$\frac{d}{dt} \binom{t}{h} \bigg|_{t=0} = \frac{(-1)^{h-1}}{h} \quad (h \ge 1)$$

give, by checking the binomial basis,

$$J_\eta(P(s+1) - P(s)) = P'(-\eta).$$

Translation of that basis also gives $J_{\eta+1}(P) = J_\eta(P(s-1))$. Consequently

$$J_{\eta+1}(P) - J_\eta(P) = -P'(-\eta - 1).$$

These signs fix both cancellation arguments below.

2. The exact rational kernel and the shift of S_A. As an identity of rational functions,

$$L_m(s) := \sum_{j=0}^{m} \frac{\omega_j}{s-j} = \frac{(-1)^m m!}{s(s-1)\cdots(s-m)}.$$

Indeed, the right side has residue ω_j at $s = j$, and the remaining polynomial part vanishes at infinity. The factor $(-1)^m$ must be retained. The polynomial F satisfies the rational identity

$$\frac{F(s+1)}{F(s)} = \frac{\prod_i (s+x_i)}{\prod_i (s+y_i)}, \qquad \frac{L_m(s+1)}{L_m(s)} = \frac{s-m}{s+1}.$$

Therefore

$$A(s+1)F(s+1)L_m(s+1) = D(s)F(s)L_m(s).$$

With $u_j = A(j)c_j$, extended by $u_0 = u_{m+1} = 0$, the same identities at nonnegative integer arguments give

$$D(j)c_j = u_{j+1} \qquad (0 \le j \le m).$$

The endpoint $j = m$ follows from $D(m) = 0$; the other cases use the binomial-weight recurrence. Since

$$S_f(s) = f(s)F(s)L_m(s) - \sum_{j=0}^{m} \frac{f(j)c_j}{s-j},$$

the two preceding equations imply, with every residue and endpoint accounted for,

$$\boxed{S_D(s) = S_A(s+1).}$$

This is a rational-function calculation whose two sides are polynomials; no division at an individual root of F is being asserted.

3. The companion relation. The rational function $A(s)F(s)L_m(s)$ can be written

$$\frac{(-1)^m m!}{\prod_i b_i!} \frac{\prod_i (s+y_i-1)_{b_i+1}}{(s-1)\cdots(s-m)}.$$

Its denominator is nonzero at $s = -\eta$, including $\eta = 0$; the factor s has already been cancelled. Since $\eta \ge \operatorname{med}(y_i - 1)$ and $\eta \le \min_i(x_i - 1)$, at least two numerator factors vanish there. Hence $(AFL_m)'(-\eta) = 0$, and differentiation of the expression for S_A gives

$$S_A'(-\eta) = \sum_{j=1}^{m} \frac{u_j}{(j+\eta)^2}.$$

On the other hand,

$$\sum_{j=0}^{m} (A(j) - D(j))c_j H_{j+\eta}^{(2)} = \sum_{j=1}^{m} \frac{u_j}{(j+\eta)^2},$$

by finite telescoping. The polynomial contribution is $J_\eta(S_A - S_D) = -S_A'(-\eta)$, and the second component also telescopes. Thus

$$\boxed{\ell_\eta(Q) = 0.}$$

Only this degree-three relation is needed when multiplying the basis $1, s, s^2$ by s. It is not necessary, and has not been claimed, that $\ell_\eta(fQ) = 0$ for every polynomial f.

4. Independence of the admissible shift. If both η and $\eta+1$ lie in I_w, subtracting the first components yields

$$\ell_{\eta+1}(f)_1 - \ell_\eta(f)_1 = \sum_j \frac{f(j)c_j}{(j+\eta+1)^2} - S'_f(-\eta-1) = -(fFL_m)'(-\eta-1).$$

At $s=-\eta-1$, the function L_m has no pole. At least two of the factors of F vanish: their integer roots cover the intervals $y_i,\dots,x_i-1$, and $\eta+1\geq \mathrm{med}(y_i)$ and $\eta+1\leq \min_i(x_i-1)$. The derivative is therefore zero. The second component contains no η. It follows that $\ell_\eta(f)$ is independent of $\eta\in I_w$.

5. The three types of contiguous move. Write

$$T_w=\begin{pmatrix}0&0&-a_0/a_3\\1&0&-a_1/a_3\\0&1&-a_2/a_3\end{pmatrix},\qquad Y_w=\frac{2}{C}(\ell_\eta(1),\ell_\eta(s),\ell_\eta(s^2)).$$

The companion relation proves that right multiplication by T_w replaces the three columns by $2\ell_\eta(s)/C, 2\ell_\eta(s^2)/C, 2\ell_\eta(s^3)/C$. For an upper increase $x_i\mapsto x_i+1$,

$$F_{\mathrm{new}}(s)=\frac{s+x_i}{b_i+1}F(s),\qquad C_{\mathrm{new}}=\frac{x_i}{b_i+1}C.$$

The definitions are linear in fF, so at any common admissible shift

$$\ell_{\mathrm{new}}(f)=\frac{1}{b_i+1}\ell_{\mathrm{old}}((s+x_i)f),\qquad Y_{\mathrm{new}}=Y_w(I+T_w/x_i).$$

For a lower decrease $y_i\mapsto y_i-1$, with $y_i\geq 2$, the same calculation replaces x_i by y_i-1. A lower increase is its inverse at the target, whenever that inverse exists. The admissible shift interval expands under an upper increase or lower decrease, so the two intervals have a common element; shift independence then removes the choice.

For the negative upper parameter, let the subscript on ℓ_m record m while F, C, η stay fixed. When $m\geq 1$,

$$\omega_j(j-m)=-m(-1)^j\binom{m-1}{j},\qquad \sum_{j=0}^{m}\omega_j=0.$$

In a divided difference the identity

$$\frac{(s-m)f(s)F(s)-(j-m)f(j)F(j)}{s-j}=(j-m)\frac{f(s)F(s)-f(j)F(j)}{s-j}+f(s)F(s)$$

therefore implies

$$S_{m,F}[(s-m)f]=-mS_{m-1,F}[f],\qquad \ell_m((s-m)f)=-m\ell_{m-1}(f).$$

Consequently

$$Y_{m-1}=Y_m(I-T_m/m),\qquad Y_m=Y_{m-1}(I-T_m/m)^{-1}$$

when the inverse is defined. This is exactly the native decrease $x_3=-(m-1)\mapsto -m$, evaluated at the target.

6. Identification with the actual row state. At $(1,1,1,0;1,1,1)$, one has $F = C = 1$, $m = \eta = 0$ and

$$S_{f=1} = 0, \quad S_{f=s} = 1, \quad S_{f=s^2} = s, \qquad J_0(1) = 1, \quad J_0(s) = -\tfrac{1}{2}.$$

Thus $Y_w = \left(\begin{smallmatrix} 0 & 2 & -1 \\ 2 & 0 & 0 \end{smallmatrix}\right) = R$. Induction along any regular admissible path from this initial vertex, using the generator equalities just established, gives $Y_w = RW$. In particular, the formula applies along the stated four-block paths provided all intermediate vertices satisfy the displayed domain and inverse conditions. The argument is uniform in the integer parameters; the previously recorded finite checks are supporting diagnostics, not its logical justification. This algebraic identification does not establish the additional prime-shell divisibility or any analytic irrationality-measure estimate.

7. Domain check for the present trajectory. For $t = 3$ or 4, use the four blocks

$$(1,1,1,0;0,0,0)^{tN}, \quad (0,0,0,0;2,1,3)^{N}, \quad (10-t, 9-t, 11-t, 0; 0,0,0)^{N}, \quad (0,0,0,-1;0,0,0)^{11N},$$

where the superscript denotes the number of block steps and every elementary move is evaluated at its current vertex. During the lower-parameter block, every old lower value is at most $3N$, whereas the three positive upper values are $tN + 1 \geq 3N + 1$. Hence the lower inverses are nonsingular. All positive upper values are at least every lower value throughout the four blocks, so $b_i \geq 0$ and the shift interval is nonempty even at intermediate coordinate moves. Also

$$a_3 = m - \sum_i b_i - 3.$$

It is negative in the first three blocks, where $m = 0$, and in the last block, where $\sum_i b_i = 24N$ and $0 \leq m \leq 11N$. For the final inverse at $m \geq 1$, its determinant is

$$\det(I - T_m/m) = \frac{Q_m(m)}{a_3 m^3} = \frac{\prod_i (m + y_i - 1)}{a_3 m^2} \neq 0.$$

The endpoint is therefore covered by the preceding uniform induction:

$$w = (10N+1, 9N+1, 11N+1, -11N; 2N+1, N+1, 3N+1), \quad 2N \leq \eta \leq 9N.$$

Here $F(s) = \binom{s+10N}{8N}\binom{s+9N}{8N}\binom{s+11N}{8N}$. This establishes the finite formula for the two specified native rows on this trajectory, independently of any additional prime-shell argument.

B Baseline arithmetic

A baseline arithmetic lemma, with no extra-shell assumption. Let $N \geq 1$, $m = 11N$, $r \in \{0,1,2\}$, $E_r = 13N + r$, and

$$F(s) = \binom{s+10N}{8N}\binom{s+9N}{8N}\binom{s+11N}{8N}, \qquad c_j = (-1)^j \binom{m}{j} F(j).$$

Put

$$S_r(s) = \sum_{j=0}^{m} (-1)^j \binom{m}{j} \frac{s^r F(s) - j^r F(j)}{s - j}, \qquad J_\eta(P) = \sum_{t=0}^{\deg P} \frac{(-1)^t}{t+1} \Delta^t P(-\eta).$$

As usual $0^0 = 1$. The finite CMF readout identity established in Appendix A is

$$C_N RW(N) e_r = 2 \begin{pmatrix} \sum_{j=0}^{m} j^r c_j H^{(2)}_{j+2N} + J_{2N}(S_r) \\ \sum_{j=0}^{m} j^r c_j \end{pmatrix}. \tag{F}$$

The argument below proves arithmetic statements about the displayed finite sums; its application to the matrix walk uses (F).

Set $D_{N,r} = d_{12N} d_{E_r}$ and define ρ, ϕ, G_N as defined in the main text. Then

$$\frac{D_{N,r}}{G_N} C_N RW(N) e_r \in \mathbb{Z}^2. \tag{B}$$

Denominator calculation. If $f(s) = \binom{s+u}{b}$ and $j \in \mathbb{Z}$, then $d_b(f(s) - f(j))/(s-j)$ is integer-valued. Indeed, the degree is at most $b-1$, and its values at $j+1, \dots, j+b$ are integers, since their nonzero denominators divide d_b. Newton interpolation at these consecutive integers proves the assertion at every integer. A telescoping product difference now implies that $d_{8N} S_r$ is integer-valued. The additional factor s^r creates no denominator. Expanding each divided difference into monomials and using $\sum_j (-1)^j \binom{m}{j} j^t = 0$ for $t < m$ gives $\deg S_r \le 24N - m + r - 1 = E_r - 1$. Thus $d_{8N} d_{E_r} J_{2N}(S_r)$ is an integer, and so is $D_{N,r} J_{2N}(S_r)$.

For the harmonic term, fix a prime p and a summand j. If the highest power $P = p^a \le j + 2N$ is at most $12N$, then $v_p(D_{N,r}) \ge 2a$, which clears every reciprocal square in that summand. Otherwise $12N < P \le j + 2N \le 13N$. We have $p^{a-1} \le P/2 \le 13N/2 < 12N$, so $v_p(D_{N,r}) \ge 2a - 1$. Also $j \ge P - 2N$ and

$$8N < P, \qquad j + N \le 12N < P, \qquad P \le j + 9N \le 20N < 2P.$$

The carry at the power P in $\binom{j+9N}{8N}$ contributes a factor p. All other prime-power contributions to its valuation are nonnegative. Thus $p \mid c_j$, providing the missing factor. This proves that both entries of $D_{N,r} C_N RW(N) e_r$ are integers.

Common divisor. Fix a prime $\sqrt{50N} < p \le 12N$, and put $h = \phi(N/p)$. All factorial arguments in c_j are at most $22N < p^2$, hence

$$v_p(c_j) = \rho(N/p, j/p) \ge h. \tag{C}$$

The minimum over $0 \le y < 1$ suffices because ρ is periodic in y. Use the sign-correct partial fraction identity

$$\sum_{j=0}^{m} \frac{(-1)^j \binom{m}{j}}{s-j} = \frac{(-1)^m m!}{\prod_{j=0}^{m}(s-j)}$$

to write

$$S_r(s) = s^r F(s) \frac{(-1)^m m!}{\prod_{j=0}^{m}(s-j)} - \sum_{j=0}^{m} \frac{j^r c_j}{s-j}. \tag{P}$$

Evaluate at the E_r consecutive nodes $s = -1, -2, \dots, -E_r$. All nonzero integer factors that occur have absolute value at most $E_r + m = 24N + r < 50N < p^2$. If $F(s) \ne 0$, counting signed consecutive factors gives exactly

$$v_p\left(F(s) \frac{m!}{\prod_{j=0}^{m}(s-j)} \right) = \rho(N/p, s/p) - 1 \ge h - 1.$$

For clarity, the factor $m!/\prod(s-j)$ contributes $\lfloor m/p \rfloor - \lfloor (m-s)/p \rfloor + \lfloor (-s-1)/p \rfloor$; for integer s this is the first carry in ρ minus one. Each nonzero binomial polynomial factor is counted by the corresponding difference of floors, also for negative arguments. If $F(s) = 0$, the first term of (P) vanishes. Every other term of (P) has valuation at least $h - 1$ by (C), because $0 < |s - j| < p^2$. Therefore $v_p(S_r(s)) \ge h - 1$ at all these nodes.

Since $d_{12N} S_r$ is integer-valued, its Newton coefficients at the consecutive nodes are integers. Since $v_p(d_{12N}) = 1$, they are all multiples of p^h. Translating an integer-valued binomial basis by an integer preserves this divisibility, so the same holds for the coefficients at $-2N$. The weights $1/(t+1)$ in

J_{2N} are cleared by d_{E_r} without losing p^h. For the harmonic sum, $D_{N,r}$ contributes p^2, all harmonic indices are less than p^2, and (C) supplies p^h. The second row is a sum of the same integer coefficients. Consequently p^h divides both normalized entries.

It remains to check $12N < p \le 13N$. Here $x = N/p \in [1/13, 1/12)$, and $\rho(x, 0) = 0$: $\lfloor 11x \rfloor = \lfloor 10x \rfloor = \lfloor 9x \rfloor = \lfloor 8x \rfloor = \lfloor 3x \rfloor = \lfloor 2x \rfloor = \lfloor x \rfloor = 0$. Every binomial carry is nonnegative, so $\phi(x) = 0$. Such primes contribute nothing to G_N. Together with the established identity (F), this completes the proof of (B).

C Integral normalizations at the block boundaries

One complete multiplication route with matching normalizations. Use the four directions

$$(1,1,1,0;0,0,0), \quad (0,0,0,0;2,1,3), \quad (7,6,8,0;0,0,0), \quad (0,0,0,-1;0,0,0)$$

for $3N, N, N, 11N$ steps, respectively, with coordinate moves within each step in the displayed order. The endpoint direction is $(10, 9, 11, -11; 2, 1, 3)$. The completed-step vertices are

$$\begin{aligned}
w_k^{(1)} &= (k+1, k+1, k+1, 0; 1, 1, 1), && 0 \le k \le 3N,\\
w_k^{(2)} &= (3N+1, 3N+1, 3N+1, 0; 2k+1, k+1, 3k+1), && 0 \le k \le N,\\
w_k^{(3)} &= (3N+7k+1, 3N+6k+1, 3N+8k+1, 0; 2N+1, N+1, 3N+1), && 0 \le k \le N,\\
w_k^{(4)} &= (10N+1, 9N+1, 11N+1, -k; 2N+1, N+1, 3N+1), && 0 \le k \le 11N.
\end{aligned}$$

Every required lower inverse exists: its old lower parameter is at most $3N$, strictly below all three positive uppers. Each required upper decrease has a negative target parameter. The coefficient $-a_3 = \sum_{i=0}^{3} x_i - \sum_{j=0}^{2}(y_j - 1)$ is positive at every intermediate vertex: it starts at three; increasing the positive uppers increases it, the complete lower block leaves it at $3N + 3$, and the final negative-upper block leaves it at $13N + 3$. At completed steps its values are $3k + 3$, $9N - 6k + 3$, $3N + 21k + 3$, and $24N - k + 3$. Thus the described route has no singular generator.

Let $Y_i(k)$ be the actual two rows transported from R along this route, so $Y_i(k+1) = Y_i(k)B_i(k)$ and $Y_{i+1}(0) = Y_i(r_i)$. At each vertex use

$$F(s) = \prod_{a=0}^{2} \binom{s + x_a - 1}{x_a - y_a}, \qquad C = F(0),$$

and the finite readout formula (F), with the current $m = -x_3$ and shift η. Write $B = \max(x_a - y_a)$, $E = \sum(x_a - y_a) - m + 2$, and $H = m + \eta$. The polynomial contribution is cleared by $d_B d_E$. If both lcm indices dominate H, they also clear the harmonic contribution. The following choices therefore keep all six entries integral and match at every boundary:

block	m	η	B	E	$L_i(k)$
1	0	0	k	$3k+2$	$d_k d_{3k+2}$ $(k > 0)$
2	0	$2k$	$3N-k$	$9N-6k+2$	$d_{3N} d_{9N-6k+2}$
3	0	$2N$	$2N+6k$	$3N+21k+2$	$d_{\max(3N,2N+6k)} d_{3N+21k+2}$
4	k	$2N$	$8N$	$24N-k+2$	$d_{\max(8N,\min(12N,k+2N))} d_{24N-k+2}$

Set $L_1(0) = 1$. In block 1 the harmonic term is zero. In block 2 both indices dominate $H = 2k$, and the first also bounds $B = 3N - k$. In block 3 the three widths are $N + 7k, 2N + 6k, 8k$; their maximum is $2N + 6k$ for $0 \le k \le N$. Both lcm indices dominate $H = 2N$.

In block 4 put $H = k + 2N$ and $E = 24N - k + 2$. We have $E \geq H$, and the first lcm index dominates the binomial width $8N$. If $H \leq 12N$, both lcm indices dominate H. Otherwise a missing reciprocal-square factor can occur only for a prime power P with $12N < P \leq H \leq 13N$. The preceding harmonic carry argument applies without alteration: for a contributing $j \leq k$ with $j + 2N \geq P$, $8N < P$, $j + N \leq 12N < P$, and $P \leq j + 9N \leq 20N < 2P$. The binomial $\binom{j+9N}{8N}$ supplies the missing prime factor. Thus $Z_i(k) = L_i(k)C_i(k)Y_i(k)$ is integral in every block.

The C factors are evaluated from the same vertex, so they agree at each boundary. The lcm factors there are, successively,

$$d_{3N}d_{9N+2}, \qquad d_{3N}d_{3N+2}, \qquad d_{8N}d_{24N+2}.$$

Therefore $Z_{i+1}(0) = Z_i(r_i)$, with $Z_1(0) = R$. The final normalization is $C_N d_{12N} d_{13N+2}$ for all six entries together. For one chosen column r, the same proof uses its degree bound $E_r = \sum(x_a - y_a) - m + r$, replacing the final second index by $13N + r$; at the initial point retain the explicit normalization one. This gives the sharper columnwise endpoint $C_N d_{12N} d_{13N+r}$.

The shifts used above are admissible: each lies between the median of the three lower offsets and the minimum positive-upper offset. They are 0, $2k$, $2N$, and $2N$ in the four blocks, respectively. All binomial widths are nonnegative. These conditions are exactly those used by the finite readout identity and its shift invariance.

D All six prime intervals

All six extra prime shells: a uniform finite-field argument. This argument uses the established finite CMF readout, its admissible-shift identity, and the baseline common-divisor lemma. It proves the full additional divisor $H_{N,r}$, rather than extrapolating finite-depth tests. Fix $r \in \{0, 1, 2\}$ and a prime p with $p^2 > 50N$ lying in one of the six column-corrected shells. Put

$$a = \lfloor 8N/p \rfloor, \quad b = 8N - ap, \qquad M = \lfloor 11N/p \rfloor, \quad d = 11N - Mp, \qquad h = \phi(N/p).$$

The integer h is the minimum carry count. The shell bounds and $p^2 > 50N$ imply $p > 50N/p > 6$, so p is odd; also $a \leq 8N/p < p$ and $M \leq 11N/p < p$; also $a + 1 < p$. Work over $\mathbb{F}_p$ and write

$$f_u(X) = \binom{X + uN}{b} \quad (u = 9, 10, 11), \qquad f = f_9 f_{10} f_{11}, \quad g(X) = \prod_{j=0}^{d} (X - j), \quad z(X) = X^p - X.$$

The binomials here are polynomials of degree $b < p$; their factorial denominators are units. For a residue $k \in \{0, \ldots, p-1\}$ define

$$e(k) = \mathbf{1}_{k>d}, \quad \lambda_u(k) = \lfloor (k + uN)/p \rfloor, \quad \delta_u(k) = \mathbf{1}_{(k+uN) \bmod p < b}, \qquad \chi(k) = e(k) + \sum_u \delta_u(k).$$

Then $\chi(k) = \rho(N/p, k/p) \geq h$. Each f_u has a simple zero at k exactly when $\delta_u(k) = 1$. Consequently

$$\mathcal{Q}(X) = \frac{f(X)}{z(X)^{h-1} g(X)} \in \mathbb{F}_p[X]. \tag{Q}$$

When $h = 0$, this means $\mathcal{Q} = fz/g$, a polynomial because $g \mid z$. For $h \geq 1$, the multiplicity of each root k in the denominator is h for $k \leq d$ and $h - 1$ for $k > d$, no larger than its multiplicity in f. Also $\mathcal{Q}(k) = 0$ whenever $\chi(k) > h$.

The finite chamber table. Write $x = N/p$. On each open interval in the following table, the minimum carry is the displayed h, and every residue with $\chi(k) = h$ has the same displayed triple (e, λ, δ). The triples are ordered by $u = 9, 10, 11$.

x interval	a	M	h	e	λ	δ	$K = 3a - M + h$
$(1/8, 2/13)$	1	1	0	0	$(1,1,1)$	$(0,0,0)$	2
$(2/9, 3/13)$	1	2	2	0	$(2,2,2)$	$(1,1,0)$	3
$(3/10, 4/13)$	2	3	1	0	$(2,3,3)$	$(0,1,0)$	4
$(3/8, 5/13)$	3	4	0	0	$(3,3,4)$	$(0,0,0)$	5
$(4/9, 5/11)$	3	4	1	0	$(4,5,5)$	$(0,1,0)$	6
$(5/11, 6/13)$	3	5	2	1	$(4,5,5)$	$(0,1,0)$	6
$(8/9, 10/11)$	7	9	0	0	$(8,9,10)$	$(0,0,0)$	12
$(10/11, 12/13)$	7	10	1	1	$(8,9,10)$	$(0,0,0)$	12

(T)

These are elementary floor inequalities, and can be checked by splitting $0 \le y < 1$ at the moving boundaries $\{cx\}$ for $c \in \{11, -9, -10, -11, -1, -2, -3\}$. Their ordering changes only when cx or $(c - c')x$ is an integer. The accompanying exact rational arrangement certificate checks every open chamber and every interior boundary; it comprises 390 constant floor cells, not a sampling of integer depths. The omitted splitting values $x = 5/11, 10/11$ cannot occur for an eligible prime: they force $(N, p) = (5, 11)$ or $(10, 11)$ after reduction, and both violate $p^2 > 50N$.

For the selected column, its shell condition is $13N + r < Kp$. Thus

$$\deg(X^r \mathcal{Q}) = 13N + r - (K-1)p - 1 < p - 1, \qquad \sum_{k \in \mathbb{F}_p} k^r \mathcal{Q}(k) = 0. \tag{V}$$

The last identity is the finite-field sum of a polynomial of degree less than $p - 1$, including its constant term.

Local factorization of the exact rational kernel. Let

$$F(s) = \prod_{u=9,10,11} \binom{s + uN}{8N}, \qquad L(s) = \frac{(-1)^{11N}(11N)!}{\prod_{j=0}^{11N}(s-j)}, \qquad R(s) = F(s)L(s).$$

On the support $\chi(k) = h$, take e, λ, δ from (T) and set

$$P(V) = \prod_u (a+1)^{\delta_u} \binom{\lambda_u + V}{a + \delta_u}, \quad D_e(V) = \prod_{l=0}^{M-e} (V - l), \quad U(V) = \frac{P(V)}{D_e(V)}, \quad \kappa = (-1)^{d-M+h-1} M! d!.$$

These are independent of k. The factors in D_e are distinct because $M < p$. In the local coordinate $s = k + pV$, direct separation of the factors divisible by p gives

$$p^{1-h} R(k + pV) \equiv \kappa \mathcal{Q}(k) U(V) \pmod{p}. \tag{R}$$

Here and below congruences of rational functions are taken away from $V = 0, \dots, M - e$ before the removable factors are cancelled. Here are the details of the factor count, including the case $e = 1$. For each binomial, if $\delta_u = 0$ its normalized leading term is $f_u(k)\binom{\lambda_u + V}{a}$. If $\delta_u = 1$, division by one p gives

$$f'_u(k)(a+1)\binom{\lambda_u + V}{a+1}.$$

This follows by splitting the a or $a + 1$ numerator factors divisible by p; equivalently it is the first nonzero Lucas congruence. It remains valid when an individual high-digit binomial vanishes. For L,

the p-factors in its denominator number $M+1-e$. After their removal the remaining denominator is $(-1)^M g'(k)$ if $e=0$, and $(-1)^M g(k)$ if $e=1$. The numerator unit is $(-1)^d M!d!$. Finally the product of the low-digit f_u values or derivatives, divided by $g'(k)$ or $g(k)$, is $(-1)^{h-1}\mathcal{Q}(k)$ by (Q). These identities prove (R). For $\chi(k)>h$, the left side has zero leading term and $\mathcal{Q}(k)=0$, so (R) continues to hold with zero right side. All factor counts are exact at the first p-power level because $p^2>50N$ bounds the relevant original factorial arguments.

Let $A_l=\operatorname{res}_{V=l}U(V)$, and let $B(V)$ be the polynomial part of U:

$$U(V)=B(V)+\sum_{l=0}^{M-e}\frac{A_l}{V-l}.$$

Set $c_j=0$ for indices outside $0\le j\le 11N$. Taking residues in (R), using $R(s)=S_0(s)+\sum_j c_j/(s-j)$, yields

$$p^{-h}c_{k+lp}\equiv\kappa\mathcal{Q}(k)A_l \pmod p. \tag{C'}$$

For residue classes outside the minimal-carry support, every c_{k+lp}/p^h vanishes modulo p, including any extra endpoint term $l=M$ when the supporting pattern has $e=1$. For a general column, the exact kernel identity is

$$S_r(s)=s^rR(s)-\sum_{j=0}^{11N}\frac{j^rc_j}{s-j}.$$

Multiplying by p^{1-h} and subtracting the congruent-residue terms from (R) now gives

$$p^{1-h}S_r(k+pV)\equiv\kappa k^r\mathcal{Q}(k)B(V) \pmod p. \tag{S'}$$

Noncongruent terms in the residue sum vanish modulo p, since every c_j is divisible by p^h. This kernel argument is necessary when $e=1$: termwise divided differences of F would overlook contributing noncongruent terms. Both sides of (S') are polynomials after subtraction, so the identity includes the derivative nodes $V=l$. More explicitly, the baseline interpolation argument established that $T_r=p^{1-h}S_r$ is integer-valued over $\mathbb{Z}_{(p)}$. Its degree is below p^2; substituting $s=k+pV$ into its Newton basis therefore gives coefficients in $\mathbb{Z}_{(p)}$ (the numerator contains at least $\lfloor t/p\rfloor$ factors of p for the basis element of degree t). Thus reduction and cancellation at these nodes are legitimate.

An admissible prime-dependent readout. Choose $\eta=Ap$, where $A=1$ if $p\ge 2N$, and $A=2$ otherwise. Every shell prime satisfies $2N\le\eta\le 9N$, so this is an allowed readout of the same matrix endpoint. For $j=k+lp$, the multiples of p not exceeding $j+\eta$ are exactly $p,\dots,(l+A)p$. The residue of p^{2-h} times the harmonic part is therefore

$$\kappa\left(\sum_l A_l\sum_{t=1}^{l+A}t^{-2}\right)\sum_{k\in\mathbb{F}_p}k^r\mathcal{Q}(k)=0.$$

The integers $1,\dots,l+A$ are units modulo p, because the harmonic indices are below p^2. Thus the harmonic part has valuation at least $h-1$.

For the polynomial part, write $T_r=p^{1-h}S_r$. The only weights in $J_\eta(T_r)$ possibly divisible by p have index $tp-1$, since $\deg S_r+1\le 13N+r<p^2$. At the Newton nodes $s=-Ap+up+k$, (S') gives $T_r(s)\equiv\kappa k^r\mathcal{Q}(k)B(u-A)$. Lucas's congruence gives

$$(-1)^{tp-1-(up+k)}\binom{tp-1}{up+k}\equiv(-1)^{t-1-u}\binom{t-1}{u} \pmod p.$$

Consequently each $\Delta^{tp-1}T_r(-Ap)$ is a constant multiple of $\sum_k k^r \mathcal{Q}(k)$, and is zero modulo p by (V). All weights in $J_\eta(T_r)$ are therefore cleared without losing a factor of p: $J_\eta(T_r) \in \mathbb{Z}_{(p)}$. The polynomial part, too, has valuation at least $h-1$.

The first entry of $C_N RW(N)e_r$ has valuation at least $h-1$; the second is a sum of c_j and has valuation at least h. Since $v_p(d_{12N}d_{13N+r}) = 2$ and $v_p(G_N) = h$, both entries of $C_N d_{12N}d_{13N+r}RW(N)e_r/G_N$ are divisible by p. Taking the product over all eligible primes in the six shells proves

$$\boxed{\frac{C_N d_{12N}d_{13N+r}}{G_N H_{N,r}} RW(N)e_r \in \mathbb{Z}^2.}$$

The analytic section supplies the growth, error, and independence arguments needed to turn this arithmetic statement into an irrationality-measure bound.

E A reproducible rational enclosure for the arithmetic rate

The moving breakpoints in the minimization over y are $\{sx\}$, for $s \in \{11, -10, -9, -11, -2, -1, -3\}$, together with 0 and 1. Their cyclic ordering can change only when sx or $(s-t)x$ is an integer. The separate floor $\lfloor 8x \rfloor$ changes only when $8x$ is integral. Thus it suffices to partition $[0,1]$ at every k/c for

$$c \in \{1, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 14, 20, 21, 22\}.$$

On each resulting open cell the minimum is constant: no breakpoint crossing or individual floor change can occur. At a sample x the minimum is obtained by checking the y breakpoints themselves and one point in each gap. Values at the isolated x boundaries do not affect the integral. There are 94 cells; the exact program computes their heights h_i and endpoints a_i, b_i. It gives

$$M := \int_0^1 \phi(x)\,dx = \frac{3517}{4095}.$$

For a positive-height cell and integer $k \geq 0$, its contribution is $h_i((k+a_i)^{-1} - (k+b_i)^{-1})$. The program sums the first $K = 2000$ periods, rounding each positive term down to a multiple of 10^{-15}. There are 106000 such terms; their rounded sum is $12127998763267375/10^{15}$. Periodicity and $k \leq k+x \leq k+1$ give the tail enclosure

$$\frac{M}{K+1} \leq \int_K^\infty \frac{\phi(x)}{x^2}\,dx \leq \frac{M}{K-1}.$$

Adding the rounding error and the tail yields

$$\frac{159010385641752609841}{131105520000000000000} \leq \gamma \leq \frac{158851460347347083647}{130974480000000000000}.$$

Numerically these endpoints are 12.12842797479104 and 12.128428404323277. In particular

$$25 - \gamma - \frac{233}{120} \leq \frac{143297092558247390159}{131105520000000000000} < \frac{10929906}{1000000}.$$

The final inequality is checked by integer cross multiplication in the supplied Python program. The finite partition and directed summation are implemented with Python's exact `Fraction` type and integer division; their uniform reduction is the argument above, and their exhaustive evaluation is in `rate_certificate.py`. The proof uses these exact finite computations and the displayed uniform reductions.

F Short code accompanying the claims

The full scripts, dependency pin, results, and commands are in `code/`. The core RamanujanTools loop is reproduced here. It checks finite native multiplication identities supporting the uniform generator argument.

```
from ramanujantools import Position
from ramanujantools.cmf.pfq import pFq
import sympy as sp
from flint import fmpq, fmpq_mat

axes = sp.symbols('x0 x1 x2 x3 y0 y1 y2')
cmf = pFq(4, 3, 1)

def ordered_walk(N, t=3):
    w = [1, 1, 1, 0, 1, 1, 1]
    W = fmpq_mat([[1,0,0], [0,1,0], [0,0,1]])
    blocks = [(t*N, (1,1,1,0,0,0,0)),
              (N, (0,0,0,0,2,1,3)),
              (N, (10-t,9-t,11-t,0,0,0,0)),
              (11*N, (0,0,0,-1,0,0,0))]
    for count, direction in blocks:
        for _ in range(count):
            for i, amount in enumerate(direction):
                for _ in range(abs(amount)):
                    E = cmf.M(axes[i], amount >= 0).subs(
                        Position(dict(zip(axes, w))))
                    E = fmpq_mat([[fmpq(str(E[a,b]))
                          for b in range(3)] for a in range(3)])
                    W = W * E
                    w[i] += 1 if amount > 0 else -1
    return W

for N in (1, 2, 3):
    assert ordered_walk(N, 3) == ordered_walk(N, 4)
```

Run `python run_checks.py` in the accompanying package to execute the exact certificates and the finite diagnostics. The README describes each program and its role. No proof-assistant files are included.

References

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